Maternal-fetal communication of circadian phase in a precocious rodent, the spiny mouse

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WEAVER, DAVID R., AND STEVEN M. REPPERT. *Maternal-fetal communication of circadian phase in a precocious rodent, the spiny mouse.* Am. J. Physiol. 253 (Endocrinol. Metab. 16): E401–E409, 1987.—The development of circadian rhythms was examined in a precocious rodent species, the spiny mouse. Spiny mouse pups born and reared in constant darkness expressed robust circadian rhythms in locomotor activity as early as *day 5* of life. Free-running activity rhythms of pups born and reared in constant darkness were coordinated with the dam on the day of birth. Postnatal maternal influences on pup rhythmicity are minimal in this species, as pups fostered on the day of birth to dams whose circadian phases were opposite to the pups' original dams were coordinated with their original dams on the day of birth. Studies using 2-deoxy-D-[1-^{14}C]-glucose autoradiography showed that there were synchronous (coordinated) rhythms in metabolic activity in the maternal and fetal suprachiasmatic nuclei, directly demonstrating prenatal coordination of maternal and fetal rhythmicity. Maternal-fetal coordination of circadian phase was not the result of direct entrainment of the fetuses to the environmental light-dark cycle. These results demonstrate that there is prenatal communication of circadian phase in this precocious species, without demonstrable postnatal maternal influences on pup circadian rhythmicity. Spiny mice therefore represent an important animal model in which circadian rhythms in the postnatal period can be used to precisely assess prenatal influences on circadian phase.

Acomys cahirinus; circadian rhythms; prenatal entrainment; suprachiasmatic nuclei

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CIRCADIAN RHYTHMS are endogenous biological rhythms with a cycle length of ~24 h. Extensive studies in rodents have established that these rhythms are generated by a circadian pacemaker (biological clock) in the suprachiasmatic nuclei (SCN) of the anterior hypothalamus (13). This biological clock is coordinated (entrained) to the environmental light-dark cycle primarily by the monosynaptic retinohypothalamic pathway (13).

Most overt circadian rhythms in physiology and behavior are first expressed during the 2nd and 3rd wk after birth in rats (1). However, a biological clock responsible for the generation of circadian rhythms begins to function earlier in development. Perinatal SCN function has been suggested by monitoring a variety of circadian rhythms under constant conditions during the postnatal period (2, 3, 8, 16, 27, 29). Prenatal SCN function has been shown directly by demonstrating rhythms in metabolic activity and vasopressin messenger RNA levels in the SCN of fetal rats (18–22). These fetal rhythms are not directly entrained in utero by the environmental light-dark cycle, but are coordinated to it by the dams' circadian timing system, demonstrating that there is maternal entrainment of the developing animal (18). This maternal entrainment occurs at a time when the developing animal is incapable of direct entrainment to light-dark cycles (4, 7, 12, 18, 25, 28).

Maternal influences during the postnatal period can also affect pup circadian phase. In rats, the potency of the postnatal influence of the dam on pup phase declines with age and is quite variable (16). In some cases, postnatal maternal influences completely obscure prenatal maternal entrainment (27). For this reason, monitoring a behavioral or hormonal rhythm during the postnatal period is not always a valid method to examine prenatal influences on the developing circadian system. This is especially true of altricial species, in which the pups are dependent on their mothers for an extended period during the postnatal period. In these species, postnatal maternal influences on pup rhythmicity are necessary to keep the pups synchronized to each other and to the dam until the pups become responsive to environmental lighting (17).

In precocious species, on the other hand, the pups are more independent, becoming responsive to the environment before or shortly after birth. In one precocious rodent species, the spiny mouse, the pups are mobile and visual, auditory, and olfactory senses are functional within hours of birth (see Ref. 15; Weaver and Reppert, unpublished observations). The pups are exposed (and presumably responsive) to the environment during the immediate postnatal period, because the dam does not build a nest and the pups' eyes open shortly after birth. In this species, therefore, there appears to be little need for postnatal maternal influences on pup circadian phase. If this were the case, then spiny mice may represent an important animal model in which postnatal behavioral rhythmicity can be used to precisely assess prenatal influences on circadian phase in the developing animal.

In the present series of experiments, we examine pre- and postnatal maternal influences on pup circadian rhythmicity in spiny mice to *1*) investigate the existence of prenatal maternal influences on pup circadian rhythmicity in this species and *2*) test our hypothesis that postnatal maternal influences would be less apparent in a precocious species.

MATERIALS AND METHODS

Animals. The spiny mice (*Acomys cahirinus*) used to establish our breeding colony were generously donated by Dr. Betsy Peitz, Department of Biology, California State University, Los Angeles. Our breeding colony of spiny mice was maintained in harems of four to eight animals in polycarbonate cages (48 × 27 × 20 cm) contained within well-ventilated environmental compartments in which the light-dark cycles were controlled automatically. The animal room was temperature controlled (21 ± 1°C) and humidity controlled (50%). One set of harems was maintained in a diurnal light-dark cycle (LD, lights on from 0400–1800 daily), while another was kept in a reversed light-dark cycle (DL, lights on from 1600–0600 daily). The light portion of the light-dark cycles consisted of white light from four 20-W cool white fluorescent bulbs, which were automatically controlled by Tork timers. The approximate light intensity at midcage level was 250 $\mu W/cm^2$. Dim red light (wavelength >620 nm, 0.8 $\mu W/cm^2$) from special fluorescent tubes (Litho light no. 2, 20 W, Chemical Products, N. Warren, PA) was provided constantly, allowing animals to be examined at night. Dim red light was also present during periods referred to as constant darkness. Food (sunflower seeds and Purina cat chow) and water were always available.

For monitoring locomotor activity, animals were placed in individual cages equipped with running wheels. All pregnant dams were housed in individual cages before parturition, and each dam and her pups remained together until the time of weaning (generally on *day 13*, with birth = *day 0*).

Spiny mice have a gestation length of 38–40 days, and pregnancy can be detected by palpation as early as gestational *day 20*. Palpation was used to determine pregnancy and to estimate the approximate stage of gestation.

Pregnant spiny mice were blinded by bilateral orbital enucleation under methoxyflurane anesthesia (Metofane, Pitman-Moore, Washington Crossing, NJ).

The experimental procedures employed in this study have been approved by the Massachusetts General Hospital Subcommittee on Animal Care (Institutional Animal Care and Use Committee).

Running-wheel activity. Animals were housed individually while monitoring locomotor activity except in the case of dams with pups. A running wheel (11.5 cm diam) was attached to the cage lid, and two magnetic actuators (Hamlin no. 5805, Lake Mills, WI) were attached to opposite sides of the wheel. The wheels were lined with window screen to further facilitate rotation of the wheel by pregnant dams and young offspring. With each half-rotation of the wheel, a magnetic actuator attached to the wheel passed a fixed proximity sensor (Hamlin no. 5804) and resulted in a switch closure within the proximity sensor. The proximity sensor was wired to one of two systems that recorded the occurrence of switch closures.

In our initial studies, each switch closure resulted in a pen deflection of a channel on an Esterline-Angus event recorder (model A620X, Indianapolis, IN). The records were manually processed into actograms by cutting the record into strips representing 24-h periods for each channel and arranging these strips with successive days below the first 24-h period (Figs. 1 and 2). In subsequent studies, the number of switch closures per 10-min interval was recorded and stored automatically on floppy disk by an automated, IBM computer-based data collection system (Dataquest III, Minimitter, Sunriver, OR). Stored data were retrieved using another subroutine and printed in the form of actograms after log transformation of the number of switch closures per 10-min interval.

The phase of activity onset in each record was determined by drawing an eye-fitted line through the daily activity onsets for a 3-wk period (14). For pups, the line through activity onsets in the postnatal period was extrapolated to the day of birth to estimate the time of activity onset on the day of birth (Fig. 2). From this information, the dam-pup phase difference on the day of birth (*day 0*) and at weaning (*day 13*) was calculated for each pup.

Deoxyglucose autoradiography. For studies of the metabolic activity of the SCN, 2-deoxy-D-[1-^{14}C]glucose (2DG, 12.5 μCi/animal, sp act 59 mCi/mmol, Amersham, Arlington Heights, IL)) was administered by intracardiac injection to manually restrained, pregnant spiny mice during the last week of gestation. Dams were placed into constant darkness 12–36 h before 2DG injection. Forty-five minutes after injection, each dam was killed by decapitation and the room lights were turned on; the fetuses were then removed from the uterus and decapitated. Maternal and fetal brains were carefully dissected from the skulls and frozen in cooled 2-methylbutane (−20 to −30°C), covered with embedding matrix (Lipshaw M-1, Detroit, MI), and stored at −80°C. Coronal sections (20 μm) through the anterior hypothalamus were cut on a cryostat at −20°C, thaw mounted on cover slips, and dried on a slide warmer. Autoradiographs were generated by exposing the sections to Kodak SB-5 film for 5–7 days. After film exposure, the sections were stained with cresyl violet, mounted on slides, and examined by light microscopy to determine the location of the SCN.

Optical density (OD) measurements were obtained using Drexel University Image Processing Center "Brain Software Package" run on an IBM AT computer with a Circon MV 9015-H monochrome microvideo camera. The optical densities determined were always within the linear portion of the optical density vs. gray level standard curve. The relative metabolic activity of the SCN in each dam and fetus was determined by calculating the relative OD of the SCN. The location of the SCN was determined using the cresyl violet-stained sections, and the OD of the corresponding portion of the autoradiograph was determined. For each animal, the ODs of the SCN were measured over four to six adjacent sections, and an average relative OD was calculated as the average OD of SCN/average OD of adjacent hypothalamus. The use of relative OD allows comparisons between animals and is a form of internal control to correct for differences in the amount of DG reaching the fetuses, section thickness, and exposure time (24).

Experimental protocols. Details of the procedures for each experiment appear in RESULTS. Statistical tests (26) were performed with the criterion for significance set at $P < 0.05$.

RESULTS

Experiment 1. Development of circadian rhythmicity in locomotor activity. Because spiny mouse pups are precocious, our first objective was to determine when the pups first express circadian rhythms in locomotor activity. Pregnant spiny mice from LD were placed in constant darkness several days before birth, and their running-wheel behavior was monitored. The pups were born and reared in constant darkness and weaned at either 7 or 13 days of age into individual cages to monitor running-wheel behavior.

The animals weaned on *day 7* exhibited prominent free-running circadian rhythms in locomotor activity (Fig. 1). However, some pups did not survive weaning at this age. All animals weaned on *day 13* survived and had clear free-running rhythms of activity, so this age was selected for subsequent studies. Interestingly, in some cases it was possible to identify the activity of a pup running on the dam's wheel before weaning. The activity is attributed to the pup in these cases because *1*) the phase of activity is consistent with the pup's phase observed after weaning, *2*) it is sufficiently out of phase with the dam, and *3*) the activity attributed to the pup is not present in the dam's record after weaning of the pup. By use of these criteria, free-running rhythms of locomotor activity have been observed in a number of cases as early as *day 5*.

Experiment 2. Perinatal coordination of circadian phase. To assess whether there is coordination of circadian phase between spiny mice dams and their offspring, running-wheel behavior of pups and dams was monitored in constant darkness. Pregnant spiny mice were placed into constant darkness when close to parturition as estimated by palpation. Births occurred 1 to 13 days after transfer to constant darkness. One group of dams (n = 4) originated in LD, while another group (n = 4) was housed in DL before transfer to constant darkness. Pups were weaned to individual cages equipped with running wheels on *day 13*; where possible, littermates were weaned at different times of day to examine whether the time of day of weaning influenced circadian phase of the pups. Locomotor activity of dams and pups was monitored in constant darkness for 3 wk after weaning.

On the day of weaning (*day 13*), the activity onset of pups differed from their dams by an average of 6 h in both LD and DL groups (Table 1 and Fig. 2). When the phase plots of the pups were extrapolated back to the day of birth, the average dam-pup difference in activity onset on the day of birth was significantly reduced to 2–3 h for both groups (Wilcoxon matched-pairs sign-rank test, $P < 0.01$, n = 9 pups in each group). Thus dams and their pups maintained in constant darkness since before birth are significantly more coordinated on the day of birth than at weaning. The time of weaning had no effect on pup phase; siblings were closely synchronized with one another at both birth and weaning, regardless of whether they were weaned at the same time or 12 h out of phase from each other. There was no correlation between the number of days in constant darkness before birth and the dam-pup difference on the day of birth (Spearman's $\rho = 0.144$, $P > 0.05$).

Experiment 3. Examination of postnatal maternal influences. To assess potential postnatal maternal influences on pup circadian rhythmicity in this species, we created a situation in which the potential pre- and postnatal maternal influences on pup rhythmicity were of opposite phase. Pregnant dams were placed into constant darkness when they were approaching parturition. Pups born in constant darkness to a dam previously entrained to

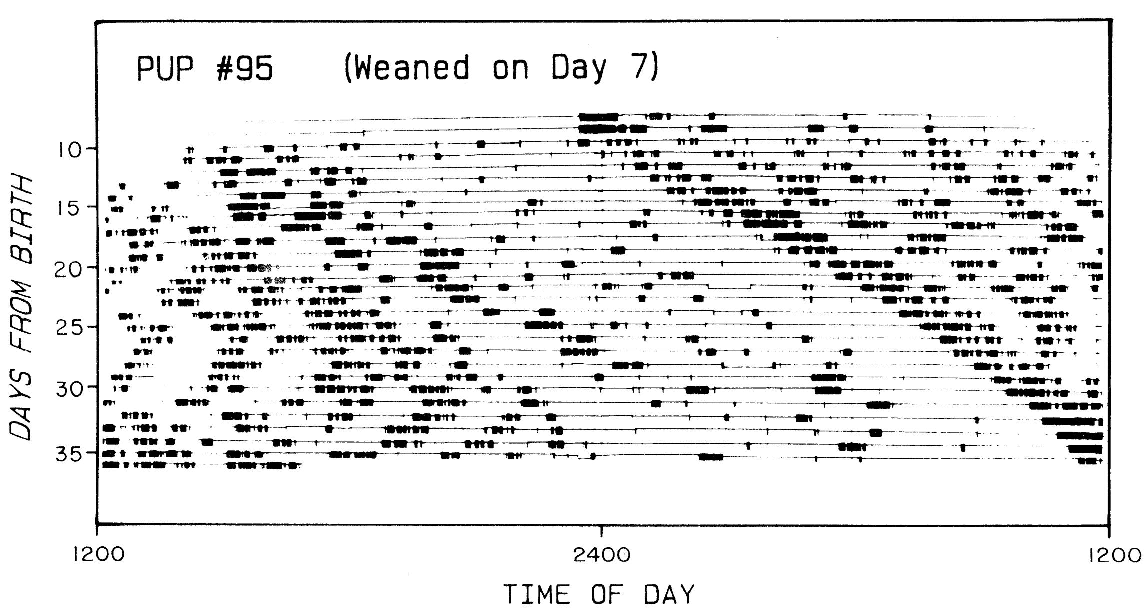

FIG. 1. Free-running circadian rhythm of locomotor activity in a spiny mouse pup born and maintained in constant darkness throughout period of monitoring. The pup was weaned from the dam to its own cage with activity wheel on *day 7* (0 = birth). Each *horizontal line* represents one 24-h period, with successive days plotted below the first. *Vertical* deflections from base line represent periods of activity.

TABLE 1. *Perinatal coordination of circadian phase*

Dam No.	Days Before Birth Into DD	Pup No.	Sex	Weaning Time	Dam-Pup Difference, h	
					Day 0	*Day 13*
LD 16	−1	*79*	Female	1525	1.5	4.5
		80	Female	1525	3.0	6.5
LD 4	−6	*81*	Female	0845	3.0	5.5
		82	Male	2200	2.0	5.0
LD 10	−2	*83*	Female	0900	2.0	4.5
		84	Male	0045	3.0	4.5
LD 25	−6	*85*	Female	0900	5.0	8.0
		86	Female	2230	0.5	7.0
		87	Male	2230	7.0	10.5
DL 15	−13	*112*	Female	2130	2.5	3.5
		113	Male	2130	5.0	9.0
DL 18	−8	*114*	Male	0900	2.0	8.0
		115	Female	0900	2.0	8.5
		116	Male	1845	0.5	8.0
DL 81	−4	*124*	ND	0930	1.5	5.0
		125	Female	1600	1.0	5.0
DL 39	−8	*128*	Female	0945	1.5	5.0
		129	Female	1945	2.5	5.0
Mean ± SE (9 pups from 4 LD dams)					3.0±0.7*	6.2±0.7
Mean ± SE (9 pups from 4 DL dams)					2.1±0.4*	6.3±0.7
Mean ± SE (18 pups from all 8 dams)					2.5±0.6	6.3±0.5

Weaning time is clock time of weaning on *day 13*. Pup phase on *day 0* was extrapolated from the phase of the free-running locomotor rhythm monitored after weaning. DD, constant darkness; LD, housed in a diurnal light-dark cycle during gestation; DL, housed in a reversed light-dark cycle during gestation; ND, not determined. * Significantly less than dam-pup difference on *day 13* (Wilcoxon matched-pairs sign-rank test, $P < 0.01$).

one lighting cycle (e.g., LD) were fostered singly on the day of birth to a dam previously entrained to the opposite lighting cycle (e.g., DL).

In each of four such foster studies, the activity onset of the fostered pup was more closely coordinated with the activity onset of the original dam than with the foster dam when the phase of the fostered pup was extrapolated back to the day of birth. The results of the two most informative foster studies (in which the dams were 12 h out of phase from one another) are presented in Fig. 3. If the foster dam had exerted a significant influence on the timing of rhythmicity in the pup, the extrapolation of the phase of the pup would not indicate such close coordination between the pup and the original dam. Thus there appears to be little or no postnatal maternal influence in this species. The synchronization of circadian phase on the day of birth therefore appears to result from prenatal entrainment of the developing circadian system.

Experiment 4. Demonstration of prenatal entrainment. 2DG autoradiography has proven useful for monitoring the rhythm in metabolic activity of the SCN in adults of several rodent species (6, 11, 23). The rhythm in SCN metabolic activity is characterized by high glucose utilization during the day and low glucose utilization at night; SCN glucose utilization can thus be used to estimate an animal's circadian time. Similarly, simultaneous assessment of the metabolic activity in the maternal and fetal SCN can be used to directly demonstrate maternal-fetal coordination of circadian phase at a time when overt circadian rhythms are not yet expressed in the developing animal (18).

Metabolic activity of the maternal and fetal SCN was examined in spiny mice using 2DG autoradiography during the last week of gestation. Pregnant LD dams were placed in constant darkness and injected 12 to 36 h later with 2DG in the middle of the subjective day (the period when the lights would have been on had the animal remained in a light-dark cycle) at 1100 or in the middle of the subjective night (the period when the lights would have been off had the animal remained in a light-dark cycle) at 2300. Autoradiographs of coronal sections through the SCN were produced, and relative ODs of the maternal and fetal SCN were determined.

Assessment of autoradiographs after injection of 2DG in the middle of the subjective day revealed that the fetal SCN had higher metabolic activity than the surrounding tissue; the SCN were visible as a pair of dark ovals on the autoradiographs (Fig. 4*A*). After subjective night injection, the fetal SCN were less metabolically active and were not discernible from surrounding tissue in the autoradiographs, although the SCN were clearly present in the corresponding cresyl violet-stained sections. There was a significant day-night rhythm in SCN relative OD in the fetuses, with ODs higher during subjective day than during subjective night (Fig. 4*B*; $P < 0.01$, Wilcoxon-Mann-Whitney two-sample test). A similar rhythm, synchronous with that in the fetuses, was apparent in the dams (subjective night relative ODs in the three dams were 0.80, 0.85, and 0.95, while relative ODs in the dams injected during subjective day were 1.08, 1.11, and 1.11). In both dams and fetuses injected during subjective day, the SCN were visible on the autoradiographs throughout the rostrocaudal extent of the SCN (defined by the corresponding cresyl violet-stained sections, 200–300 μm).

Experiment 5. Mechanism for prenatal entrainment: mother vs. environment. We next examined whether maternal-fetal coordination of phase is due to direct photic entrainment of the fetuses or due to the pups receiving a maternal entraining signal. Pregnant spiny mice were enucleated 2 to 3 wk before birth and placed into a lighting cycle opposite their circadian time; running-wheel behavior of the dams was monitored to verify that the environmental light-dark cycle was approximately opposite the dams' circadian time. After a minimum of 7 days of exposure to the out-of-phase, light-dark cycle, the blind dams were shifted to constant darkness before giving birth. The pups were born and reared in constant darkness and weaned to individual cages on *day 13*. Running-wheel behavior was monitored for 3 wk.

Pups whose dams were exposed during gestation to a light-dark cycle out of phase with maternal rhythmicity were coordinated with the dam at birth and not with the external lighting cycle (Table 2). We have also recently observed one litter in which the dam was in constant darkness throughout pregnancy, so that light could not possibly influence the pup. In this case, the single pup was coordinated to the dam on the day of birth (dam-pup difference = 2 h). These results indicate that prenatal entrainment of the fetuses is the result of maternal

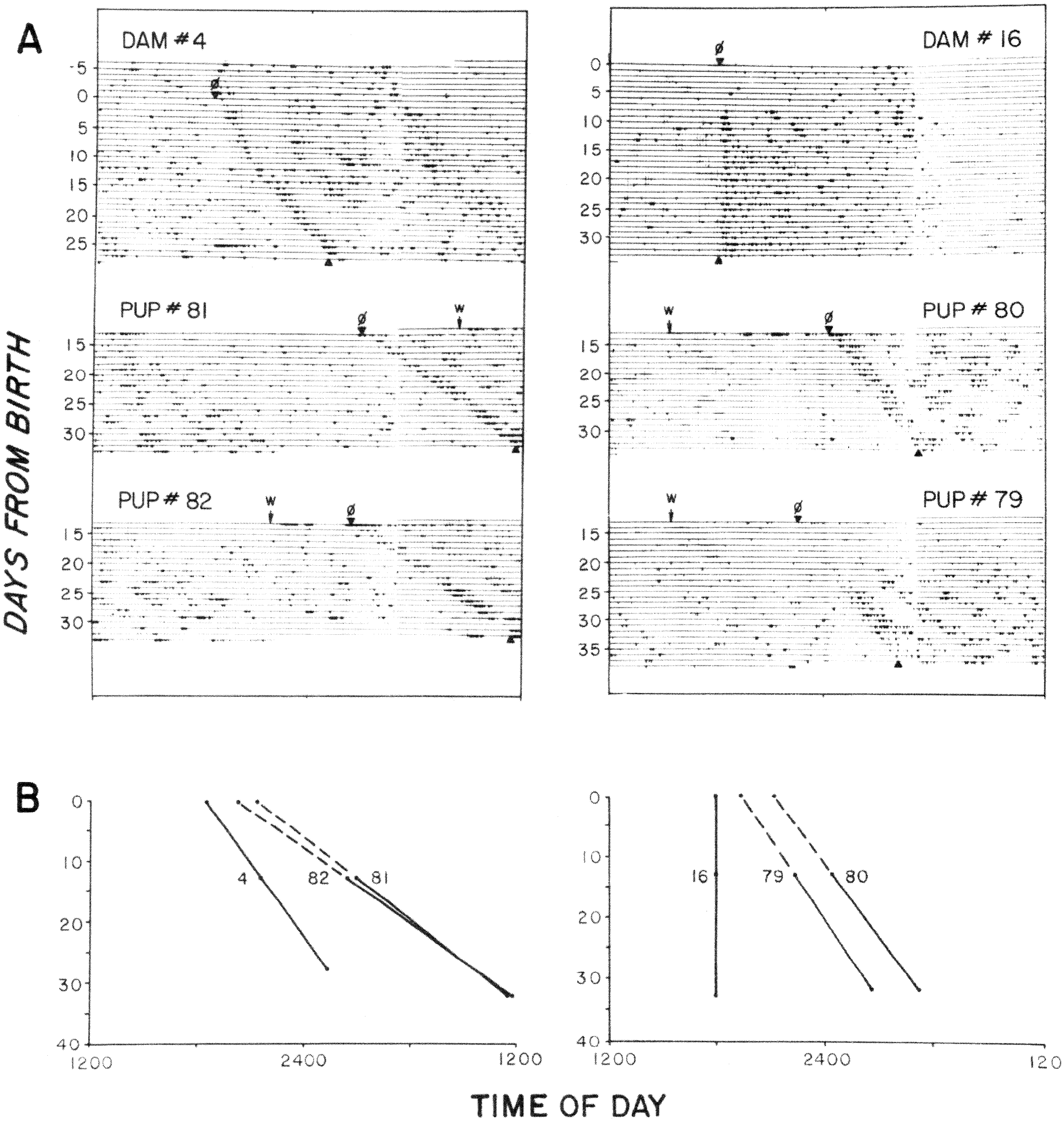

FIG. 2. *A*: free-running activity records of 2 spiny mice dams and their pups in constant darkness. Pregnant dams previously in diurnal lighting (lights on 0400–1800) were placed into constant darkness several days before birth; pups were born and raised in constant darkness. Pups were removed from the dams' cages on *day 13* (time indicated by *W* in the record) and placed in individual cages for monitoring locomotor behavior. Two points on *eye-fitted line* through activity onsets of the record are indicated by *symbol* ϕ. *B*: phase difference between dams and pups was determined by superimposition of the lines for activity onset for dams and their pups. *Solid lines* indicate areas where phase of animal was defined by data in *A*; *dashed lines* indicate extrapolation of pups' phase before weaning.

entrainment and is not due to direct entrainment of the fetus by environmental lighting, at least under the lighting conditions used in these studies.

DISCUSSION

The present experiments demonstrate maternal-fetal communication of circadian phase in a precocious rodent species, the spiny mouse. A behavioral paradigm was validated by showing that postnatal influences on pup rhythmicity are minimal in this species, allowing the use of postnatal behavior as a means to investigate prenatal circadian rhythmicity. The demonstration of synchronized rhythms of metabolic activity in the maternal and fetal SCN further indicates prenatal coordination of circadian phase. Light does not directly entrain fetal spiny mice under the conditions used, as fetuses of enucleated dams exposed to an out-of-phase lighting cycle were coordinated with the dam and not with the lighting cycle.

The behavioral experiments in this study rely on the assumption that postnatal behavioral rhythmicity can be

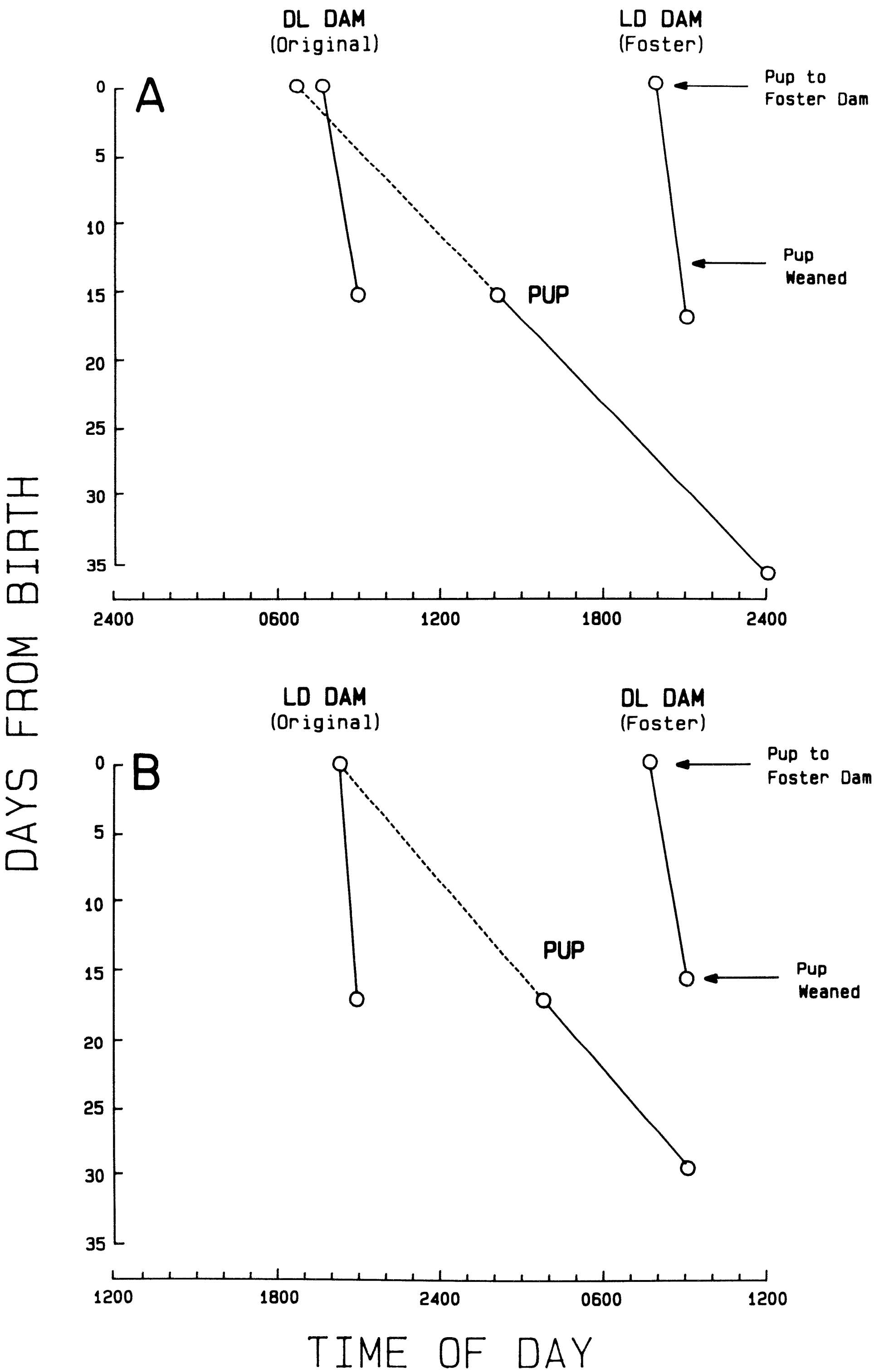

FIG. 3. *A* and *B* represent results of 2 foster experiments to assess extent of potential postnatal maternal influences on pup rhythmicity. In *A* and *B*, lines representing phase of activity onset for original and foster dams and pup fostered between them are shown. Pups fostered to a dam out of phase with original dam are clearly in phase with original dam on day of birth. LD, housed in a diurnal light-dark cycle during gestation; DL, housed in a reversed light-dark cycle during gestation.

used to infer an animal's circadian phase at earlier times. This assumption requires that *1*) extrapolation is an accurate and reliable method for determining phase at an earlier time, and *2*) events during the postnatal period do not influence pup phase.

The first of these requirements is supported by the finding that maternal-fetal coordination is also evident in pups weaned at *day 7* (Weaver and Reppert, unpublished observations), where the distance of the extrapolation is reduced. In addition, in the records of several of the pups weaned on *day 13*, it is possible to identify the activity of the pup on the dam's wheel as early as *day 5* (see *expt 1*). In these cases, the phase of the pups' observed activity was virtually identical to the phase estimated from the pups' behavior after weaning.

The second requirement, that events in the postnatal period do not influence circadian timing of the pups, is supported by the relative independence of maternal and

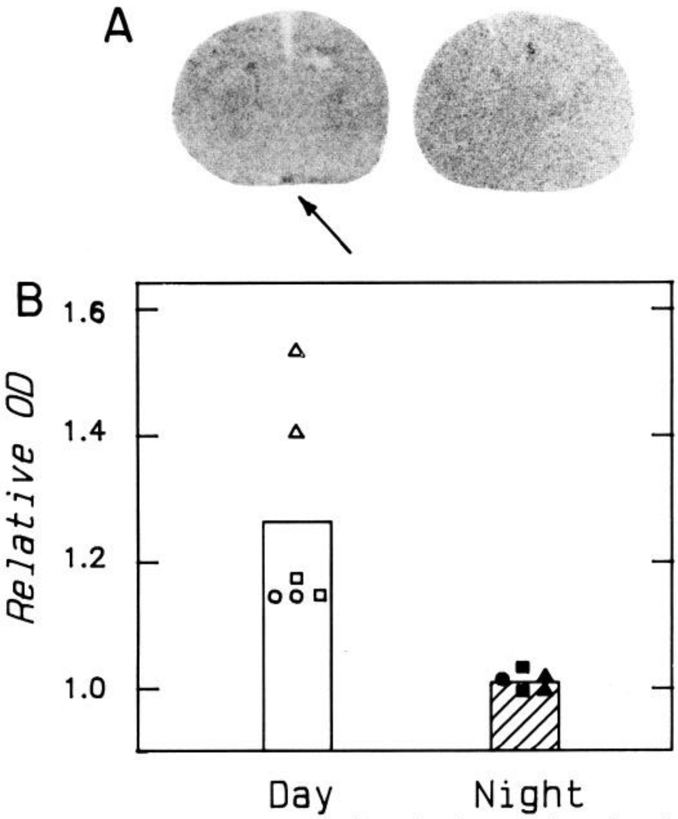

FIG. 4. *A*: representative autoradiographs from sections through midportion of fetal suprachiasmatic nuclei (SCN) after injection of 2-deoxy-D-[1-^{14}C]glucose into dam during subjective day (*left*) or subjective night (*right*). Metabolic activity of the fetal SCN (indicated by *arrow*) was higher than adjacent hypothalamus during subjective day, so SCN are readily apparent, while SCN are not visible (metabolic activity not different from surrounding tissue) during subjective night. Magnification ×4.5. *B*: relative optical density (OD) of fetal SCN from pups of dams injected during subjective day ($n = 3$) or subjective night ($n = 3$) on gestational *day 35*. For each fetus, relative OD (OD of SCN/OD of adjacent hypothalamus) was calculated for 4–6 consecutive sections through center of SCN. Relative ODs of littermates are represented by same *symbol*. Relative optical densities of the SCN for the animals whose autoradiographs appear in *A* were 1.18 and 1.00 for subjective day and night injections, respectively.

pup rhythmicity during the postnatal period. The average (mean ± SE) free-running period (cycle length) of the 13 dams in this study, 24.0 ± 0.04 h (range 23.7–24.2 h), was significantly less than the average period of the 28 pups, 24.3 ± 0.04 h (range 24.2–24.6 h; $P < 0.01$, Wilcoxon-Mann-Whitney two-sample test). In addition, foster dams of opposite circadian phase did not influence circadian rhythmicity of pups during the postnatal period (see *expt 3*). Taken together, these lines of evidence are consistent in suggesting that circadian rhythmicity in spiny mice pups is relatively free from postnatal influences and that postnatal behavior can be used to accurately estimate circadian phase at earlier developmental stages.

The onset of circadian rhythmicity in spiny mice is quite remarkable when compared with other rodent species. In rats, intrinsic SCN rhythms in metabolic activity and vasopressin mRNA levels can first be detected on gestational *days 19* and *21*, respectively (19, 22). The first overt circadian rhythm that can be demonstrated in rats is the rhythm in pineal *N*-acetyltransferase activity, which appears on postnatal *day 4* (5). Rhythms in locomotor activity, drinking behavior, and corticosterone levels appear in the 2nd and 3rd wk of life (1). In contrast, spiny mice have robust rhythms in locomotor activity as early as *day 5*. It would be interesting to determine when other circadian rhythms can first be demonstrated in this species.

Perinatal coordination of circadian phase has been demonstrated in a number of altricial rodent species (see Ref. 17 for review). In each of these species, there are significant postnatal influences on pup rhythmicity so that pups are synchronized with their dam at weaning at 3–4 wk of age. In spiny mice, however, there is no demonstrable postnatal maternal influences on pup phase. Interestingly, pups reared in constant darkness are ~6 h out of phase from their dam by *day 13*. It is unlikely that under natural conditions spiny mouse pups and their dams become desynchronized to this extent during the postnatal period. Instead, spiny mouse pups are probably exposed to the environmental light-dark cycle and become entrained to it shortly after birth. Maternally mediated prenatal entrainment of the pup may facilitate direct entrainment to the environmental light-dark cycle postnatally. Synchronization of circadian rhythmicity of the dam and her offspring during the neonatal period may be important for coordination of maternal and pup physiology and behavior. For example,

TABLE 2. *Fetuses are entrained to dam, not environment*

Dam No.	Days Before Birth		Pup No.	Sex	Weaning Time	Dam-Pup Difference, h	
	Into OPP	Into DD				*Day 0*	*Day 13*
DL *99*	−19	−6	*122*	Male	0845	3.0	8.5
			123	Male	1800	3.5	9.0
DL *117*	−14	−7	*118*	ND	0910	1.0	3.5
LD *104*	−22	−7	*126*	Female	0930, *day 8*	0.0	1.0
			127	Female	0930, *day 8*	2.0	3.0
LD *26*	−19	−8	*130*	Male	0945	1.0	7.0
			131	Female	0945	0.0	9.5
			132	Male	1945	0.0	7.0
Mean ± SE (8 pups from 4 dams)						1.3±0.5*	6.1±1.1

Weaning time is clock time of weaning on *day 13*, except as indicated. Pup phase on *day 0* was extrapolated from the phase of the free-running locomotor rhythm monitored after weaning. OPP, oppositely phased light-dark cycle; DD, constant darkness; DL, housed in a reversed light-dark cycle during gestation; LD, housed in a diurnal light-dark cycle during gestation; ND, not determined. * Significantly less than the dam-pup difference on *day 13* (Wilcoxon matched-pairs sign-rank test, $P < 0.05$).

nursing may be most efficient when maternal willingness to nurse is coordinated with the pups' interest in feeding (9).

In this study, light did not appear to directly influence circadian rhythmicity of fetuses during the prenatal period. *Experiment 5* demonstrated that maternal-fetal coordination of circadian phase was not due to direct photic entrainment of the fetus in utero under the lighting conditions used. Although it is possible that environmental lighting may influence fetal physiology if provided in sufficient intensity at a maturationally appropriate time of gestation (10), our results show that environmental lighting is not necessary for maternal-fetal coordination of rhythmicity. We conclude that maternal-fetal coordination of circadian phase is due to entrainment of the spiny mouse fetus to some maternal signal.

The nature of this maternal signal is yet to be determined; it has been most extensively studied in rats. Lesions of the maternal SCN on gestational *day 7* disrupt maternal-fetal communication of circadian phase, suggesting that some maternal circadian rhythm is the signal entraining the fetal biological clock (21). Removal (in separate experiments) of the maternal pineal, adrenal glands, ovaries, or thyroid and parathyroid glands do not disrupt the communication, however, indicating that the rhythmic secretion of hormones from these glands are not necessary (20). The ability to design studies of this sort in rats is limited, however, because either a prenatal assessment of phase must be used (e.g., 2DG autoradiography) or the pups must be fostered to an SCN-lesioned dam on the day of birth to avoid the complication of postnatal maternal influences on pup phase. Similar difficulties exist in the other altricial rodent species currently being used to study maternal-fetal communication of phase (2, 29).

The spiny mouse provides an attractive model for additional studies of the mechanisms for prenatal influences on developing circadian rhythmicity because of the unique absence of postnatal maternal influences in this species. In addition, the precision with which phase can be estimated from locomotor activity of individual animals may allow identification of more subtle prenatal influences on phase than would be possible with single time-point assessments of phase (e.g., 2DG autoradiography or pineal *N*-acetyltransferase activity). This may be particularly important if the communication of phase during the prenatal period involves multiple signals that act in concert to entrain the fetal biological clock.

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